

AIA Unit 3 Notes

1. Introduction to the Model of Cognition

1.1 What is Cognition?

Cognition refers to the mental processes that deal with knowledge. These processes encompass psychological activities that acquire, store, retrieve, transform, or apply information. Cognitions are a pervasive part of mental life, helping individuals understand and interact with the world.

Cognitive processes include:

- **Perception:** Taking in and interpreting sensory information from the environment.
- **Memory:** Storing and retrieving past experiences and learned knowledge.
- **Reasoning:** Drawing logical conclusions and making inferences.
- **Decision Making:** Evaluating options and selecting the best course of action.
- **Learning:** Updating knowledge and behaviour based on new experiences.
- **Language:** Understanding and producing human communication.

1.2 Model of Cognition

A model of cognition is a theoretical or computational representation of mental processes — such as perception, memory, and reasoning — used to understand, simulate, or predict human behaviour. These models range from simple diagrams to complex computer programs.

- **Definition & Purpose:** Designed to explain the 'inner workings' of the mind — acting as a bridge between cognitive science, psychology, and artificial intelligence.
- **Goal:** Explain how people process information, make decisions, and experience emotions in a structured, reproducible way.

Why Models of Cognition Matter for AI

Cognitive models are the blueprint AI researchers use when designing intelligent systems. By understanding how the human mind perceives, reasons, and decides, we can build machines that replicate or augment these abilities — from computer vision to natural language understanding to autonomous robots.

1.3 Cybernetics and Cognition

Cybernetics is the study of regulatory systems, feedback loops, and control mechanisms in both living organisms and machines. The 'cybernetic view' of robot cognition treats an intelligent robot as a feedback-controlled system that continuously senses its environment, processes information, and adjusts its behaviour — mirroring how biological brains operate.

Cybernetic Control Loop (applies to both brains and robots):

```

Sense Environment
↓
Process / Perceive (compare to internal model / goal)
↓
Decide Action
↓
Act on Environment
↓
Receive Feedback (observe outcome)
↓ (loop repeats continuously)

```

Key Cybernetic Principles in Robotics:

- **Feedback Control:** The robot continuously measures the gap between its current state and its goal, using that error signal to correct its actions.
- **Adaptive Behaviour:** A cognitive robot adjusts its strategy based on feedback — it does not blindly follow a fixed script.
- **Goal-Directedness:** All behaviour is oriented toward achieving a goal, just as human actions are purposeful.

1.4 Components of a Cognitive Architecture

A cognitive architecture defines the fixed structure of an intelligent system — the 'skeleton' upon which all cognitive processes run. Key components include:

Component	Description	Example
Perception Module	Takes raw sensory input and converts it into a meaningful internal representation.	Computer vision, speech recognition
World Model / Memory	Maintains an internal representation of the current state of the environment.	Map of robot's surroundings, knowledge base
Reasoning Engine	Uses the world model and goals to decide what actions to take.	Planning algorithms, inference engines
Learning Module	Updates the world model and reasoning strategies based on experience.	Reinforcement learning, neural networks
Action / Motor Control	Translates decisions into physical or digital actions.	Robot actuators, speech synthesis

2. Visual Perception

2.1 What is Visual Perception in AI?

Visual perception in AI is the capability of machines to interpret, understand, and derive actionable information from visual data like images, videos, and 3D scenes — mimicking human sight. It uses computer vision, deep learning, and sensors such as cameras and LiDAR to detect objects, recognise faces, track motion, and navigate environments.

Human vs Machine Visual Perception
Human vision: The brain interprets light signals, applies context, memory, and emotion to understand a scene holistically. AI visual perception: Algorithms detect patterns in pixel data using statistical models — powerful for specific tasks, but lacks human-like contextual understanding. It remains 'pattern matching' rather than true cognitive understanding.

2.2 Core Components and Processes

Data Input

Cameras and video feeds provide data on colour, texture, shape, and motion. The raw input is a grid of pixel values (for images) or a sequence of frames (for video).

Image Processing

Techniques like Convolutional Neural Networks (CNNs) process pixel data layer by layer — first detecting simple features (edges, corners), then progressively building up to complex patterns (shapes, objects, scenes).

Key Visual Perception Techniques

Technique	What It Does	Example
Object Detection / Classification	Identifying what objects are present in an image and where they are.	Detecting pedestrians in a self-driving car camera feed
Segmentation	Partitioning an image into meaningful segments — assigning each pixel to a region or object.	Separating road, sky, buildings in a scene
Tracking	Monitoring objects across multiple video frames over time.	Tracking a ball in sports video
3D Reconstruction	Using stereopsis (two cameras) or depth sensors to perceive depth and build a 3D model of the scene.	Mapping a room with a robot's stereo camera
Face Detection	Locating human faces in an image regardless of lighting or angle.	Camera auto-focus, access control systems

Technique	What It Does	Example
Motion Estimation	Detecting and quantifying movement between frames.	Optical flow in video surveillance

2.3 Key Applications of Visual Perception

- **Autonomous Vehicles:** Real-time detection of pedestrians, lane markings, traffic signals, and other vehicles to navigate safely.
- **Medical Imaging:** Analysing X-rays, CT scans, and MRIs for early disease detection — cancers, fractures, organ abnormalities.
- **Surveillance / Security:** Facial recognition, anomaly detection, and behaviour analysis in security camera systems.
- **Manufacturing:** Quality control — detecting product defects on production lines with greater speed and accuracy than human inspection.
- **Augmented Reality (AR):** Overlaying digital information on real-world scenes by understanding the physical environment in real time.

2.4 Challenges and Limitations

Challenge	Description
Lack of Context	AI lacks common sense — it cannot infer meaning from context the way humans naturally do.
Data Dependency	Performance is heavily dependent on the quality and diversity of training data. Poor data = poor perception.
Susceptibility to Illusions	AI can be fooled by adversarial examples — images with tiny imperceptible changes that completely confuse the model.
Lighting & Occlusion	Changes in lighting conditions, shadows, or objects partially blocking the view significantly reduce accuracy.
Computational Cost	Real-time visual perception for complex scenes requires significant processing power (GPUs).

Key quote from teacher's notes: 'While AI visual perception has advanced rapidly, it remains a pattern matching process rather than true cognitive understanding.'

3. Visual Recognition

3.1 What is Visual Recognition?

Visual recognition in AI is a computer vision technology that enables machines to identify, analyse, and interpret elements — such as objects, faces, text, and actions — within digital images or videos. It uses deep learning, particularly Convolutional Neural Networks (CNNs), to detect patterns in data — mimicking human visual perception.

Visual Perception vs Visual Recognition

Visual Perception = the broader ability to process and understand visual input (including depth, motion, 3D structure). Visual Recognition = the specific task of identifying and labelling what is in an image (objects, faces, text, actions). Recognition is a subset of perception.

3.2 Key Components and Techniques

Image Classification

Labelling an entire image with a single category based on its dominant content. The model outputs a class label for the whole image.

- **Example:** Classifying a photo as 'dog' vs 'cat', or 'chest X-ray: pneumonia' vs 'normal'.
- **Algorithm:** CNN followed by Softmax output layer.

Object Detection

Going beyond classification — locating and identifying multiple objects within a single image simultaneously. Each detected object gets a bounding box and a class label.

- **Example:** Identifying cars, pedestrians, and traffic signs all in the same image for autonomous driving.
- **Algorithms:** YOLO (You Only Look Once), SSD (Single Shot Detector), Faster R-CNN.

Facial Recognition

Identifying or verifying individuals by analysing unique facial features — the geometry of the face, distances between features, and texture patterns.

- **Identification:** 'Who is this person?' — matches face against a database.
- **Verification:** 'Is this the person they claim to be?' — confirms identity.
- **Applications:** Smartphone unlock, airport security, law enforcement, tagging in photos.

Segmentation

Partitioning an image at the pixel level — each pixel is assigned to a specific object or region class.

- **Semantic Segmentation:** Every pixel gets a class label (road, sky, person) — but individual instances are not distinguished.
- **Instance Segmentation:** Each individual object instance gets its own unique mask — even two people of the same class get separate masks.
- **Example:** Surgical robotics needs exact object boundaries, not just bounding boxes.

3.3 How Visual Recognition Works

1. **Data Collection & Annotation:** Gather a large dataset of images and label each one with the correct class or bounding box.
2. **Model Training:** Feed labelled images into a deep learning model (e.g., CNN). The model adjusts its weights through backpropagation to minimise classification error.
3. **Inference:** Apply the trained model to new, unseen images — the model outputs its prediction.

Visual Recognition Pipeline:

```
Raw Image
↓
Preprocessing (resize, normalise pixel values)
↓
CNN Feature Extraction (convolution + pooling layers)
↓
Classification / Detection Head (fully connected layers + Softmax)
↓
Output: class label, bounding box, confidence score
```

3.4 Key Applications

Domain	Application	Example
Automotive	Self-driving car perception	Detecting pedestrians, lanes, and traffic signs in real time
Healthcare	Medical image analysis	Automated detection of tumours in MRI scans
Manufacturing	Visual quality inspection	Detecting micro-defects on circuit boards
Security	Surveillance and access control	Facial recognition at airport border control
Retail	Product recognition and checkout	Amazon Go stores — identify items without scanning
Social Media	Auto photo tagging	Facebook auto-tagging friends in uploaded photos

3.5 Challenges

- **Data Quality / Quantity:** Requires large, diverse, accurately-labelled datasets — expensive and time-consuming to create.
- **Robustness:** Must maintain accuracy despite variations in lighting, camera angles, image blur, or partial occlusions.
- **Privacy Concerns:** Facial recognition raises significant ethical concerns around surveillance and data privacy.
- **Bias:** If training data is not diverse, the model may perform poorly on under-represented groups.

4. Machine Learning in the Context of Perception

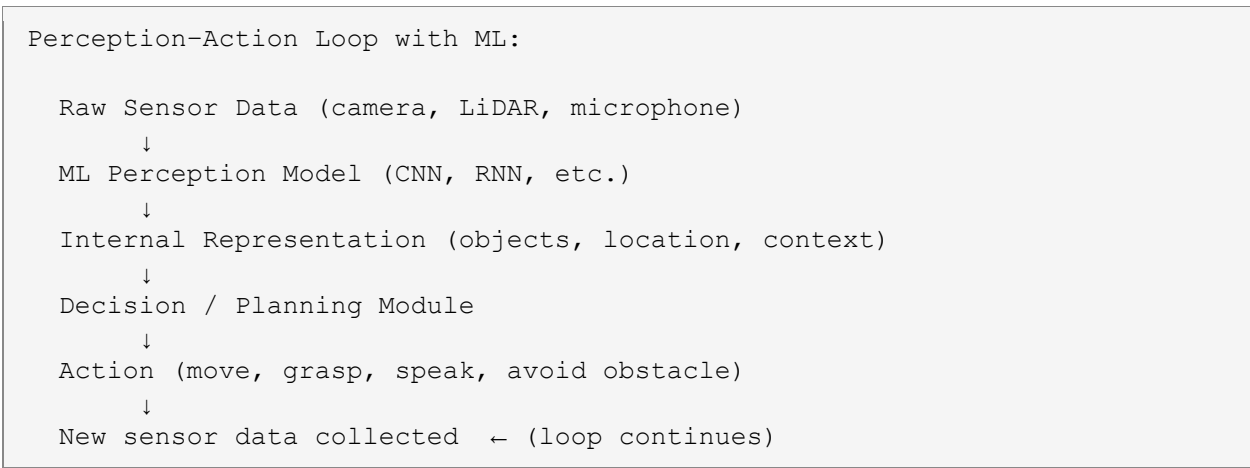
4.1 Role of Machine Learning in Robot Perception

Machine Learning is the engine that powers modern robot perception. Rather than hand-programming rules for every possible scenario a robot might encounter, ML allows the robot to learn perceptual skills directly from data — making it far more flexible and robust.

Perceptual Task	ML Technique Used	What is Learned
Object Recognition	CNN (supervised)	Which visual features indicate each object class
Scene Understanding	Semantic Segmentation (deep learning)	What each region of an image represents
Speech / Audio	Recurrent Neural Networks, Transformers	Patterns in audio waveforms that correspond to words
Navigation	Reinforcement Learning	Which actions lead the robot to its goal safely
Anomaly Detection	Unsupervised Autoencoder /	What 'normal' looks like so deviations are flagged

4.2 The Perception–Action Loop

In cognitive robotics, perception and action are tightly coupled in a continuous loop. ML models are trained to map sensory input directly to meaningful actions — this is the essence of end-to-end learning in robotics.



4.3 Why Traditional Rule-Based Systems Are Insufficient

- **Combinatorial explosion:** The number of possible visual scenes is astronomically large — impossible to write rules for every case.

- **Noise and variability:** Real-world images are noisy, blurry, and variable in lighting — rules break down quickly.
- **Adaptability:** A rule-based system cannot improve when it encounters new environments. An ML system can learn from new data.
- **Conclusion:** Machine learning — especially deep learning — is now the standard approach for robot perception because it handles complexity, noise, and variability that rule-based systems cannot.

5. Soft Computing Tools

5.1 What is Soft Computing?

Soft computing tools are AI-driven techniques designed to model complex, imprecise, and uncertain real-world problems — offering 'good enough' solutions rather than exact ones. Unlike hard computing (classical algorithms that require precise inputs and give exact outputs), soft computing tolerates imprecision and uncertainty to produce practical, useful results.

Feature	Hard Computing	Soft Computing
Data requirement	Precise, exact inputs	Approximate, uncertain, or incomplete data
Output	Exact solution	Good enough / approximate solution
Uncertainty handling	Cannot handle	Core strength — designed for uncertainty
Adaptability	Fixed algorithm	Learns and adapts from data
Examples	Sorting algorithm, SQL query	Fuzzy logic, neural networks, genetic algorithms

The main pillars of soft computing:

- Fuzzy Logic
- Artificial Neural Networks (ANN)
- Genetic Algorithms (GA)
- Support Vector Machines (SVM)
- Machine Learning & Expert Systems
- Hybrid Systems (e.g., Neuro-Fuzzy)

5.2 Fuzzy Logic

Fuzzy logic handles imprecision and partial truth. In classical (Boolean) logic, a statement is either completely TRUE (1) or completely FALSE (0). Fuzzy logic allows degrees of truth — a value between 0 and 1 — making it ideal for vague or imprecise concepts.

Classical Logic Example:

'Temperature is hot' → TRUE or FALSE

Fuzzy Logic Example:

'Temperature is hot' → 0.8 (80% degree of membership in 'hot')

'Temperature is warm' → 0.6

'Temperature is cold' → 0.1

Fuzzy Rule Example:

IF temperature is HIGH AND humidity is HIGH THEN comfort is LOW

- **Fuzzy Inference System:** Takes crisp input → fuzzifies it → applies fuzzy rules → defuzzifies output back to a crisp value.
- **Applications:** Air conditioning control, washing machines, camera autofocus, medical diagnosis, autonomous vehicle speed control.
- **Key advantage:** Can model human expert knowledge in natural language ('if speed is fast and road is wet, brake firmly').

5.3 Artificial Neural Networks (ANN) as Soft Computing

ANNs are a key soft computing tool. They mimic the human brain to learn patterns from data for tasks like pattern recognition, speech recognition, and prediction. (Covered in depth in Unit II — refer to ANN and Backpropagation sections.)

- **Soft computing role:** Handles noisy, incomplete, and high-dimensional data — produces good approximations even when exact solutions are impossible.
- **Robustness:** A network with many neurons can still function even if some neurons are damaged — similar to biological brains.

5.4 Genetic Algorithms (GA)

Genetic Algorithms are optimisation and search algorithms inspired by the process of natural selection and evolution. They evolve a population of candidate solutions toward better and better solutions over successive generations.

How a Genetic Algorithm works:

4. Initialise a population of random candidate solutions (chromosomes).
5. Evaluate each solution using a fitness function (how good is it?).
6. Select the fittest individuals to be parents for the next generation.
7. Apply crossover (combine parts of two parents to create offspring).
8. Apply mutation (randomly alter small parts of offspring).
9. Replace old population with new generation. Repeat from Step 2.

Analogy with Natural Evolution:

Population	= set of candidate solutions
Chromosome	= one candidate solution (encoded as a string)
Gene	= one parameter / decision variable
Fitness	= quality score of a solution
Selection	= survival of the fittest
Crossover	= combining two parents' genes
Mutation	= random small change to a gene
Generation	= one iteration of the algorithm

- **Applications:** Scheduling, route optimisation, neural network architecture search, game AI, engineering design.
- **Key advantage:** Can search large, complex solution spaces where traditional optimisation methods get stuck.

5.5 Support Vector Machines (SVM)

Support Vector Machines are supervised learning models used for classification and regression. They work by finding the optimal decision boundary (hyperplane) that maximally separates different classes of data.

Key Concepts:

- **Hyperplane:** A decision boundary that separates classes. In 2D: a line. In 3D: a plane. In higher dimensions: a hyperplane.
- **Support Vectors:** The data points closest to the hyperplane — they 'support' (define) the boundary. Only these points determine the hyperplane.
- **Margin:** The distance between the hyperplane and the nearest support vectors on each side. SVM maximises this margin.
- **Kernel Trick:** For non-linearly separable data, SVMs use a kernel function to map data to a higher-dimensional space where a linear boundary can be found.

SVM Decision: Which side of the hyperplane is the data point on?

Data point on side A → Class +1

Data point on side B → Class -1

Goal: Find the hyperplane that maximises the margin between the two classes.
Large margin = better generalisation = less overfitting

- **Applications:** Text classification, image recognition, bioinformatics (gene classification), handwriting recognition.
- **Advantage over Neural Networks:** Effective in high-dimensional spaces, works well with small datasets, theoretically grounded in optimisation.

5.6 Hybrid Systems — Neuro-Fuzzy

Hybrid systems combine two or more soft computing techniques to leverage the strengths of each and overcome individual weaknesses.

- **Neuro-Fuzzy Systems:** Combine neural networks (learning from data, adapts automatically) with fuzzy logic (handles uncertainty, interpretable rules). The neural network learns the fuzzy membership functions and rules from data — giving a system that is both adaptive and interpretable.
- **Example:** ANFIS (Adaptive Neuro-Fuzzy Inference System) — widely used in control systems and prediction.

- **GA + Neural Network:** Genetic algorithms can be used to optimise the architecture and weights of neural networks — avoiding manual hyperparameter tuning.

5.7 Applications of Soft Computing

Application Area	Technique Used	Specific Use Case
Control Systems	Fuzzy Logic, ANN	Adaptive industrial process control, robotics motor control
Image Processing & Pattern Recognition	ANN (CNN), SVM	Face recognition, medical image analysis, defect detection
Data Analysis & Prediction	ANN, GA	Financial forecasting, weather prediction, demand planning
Decision Making	Fuzzy Logic, Expert Systems	Medical diagnosis, risk assessment under uncertainty
Optimisation	Genetic Algorithms	Network routing, job scheduling, engineering design
Robot Perception	ANN + Fuzzy (Hybrid)	Obstacle avoidance, adaptive sensor fusion in robots

6. Robot Cognition

6.1 What is Cognitive Robotics?

Cognitive robotics is a branch of robotics that endows autonomous machines with human-like cognitive functions — reasoning, learning, perception, and decision-making — to interact intelligently with complex, dynamic, and human-centric environments. It blends artificial intelligence, cognitive science, and neuroscience to create robots capable of adapting to new situations rather than relying on pre-programmed instructions.

Cognitive robotics may be considered the engineering branch of embodied cognitive science — consisting of robotic process automation, AI, machine learning, deep learning, optical character recognition, image processing, process mining, analytics, software development, and system integration.

6.2 What is Cognition? (in the context of Robotics)

Cognitions are mental processes that deal with knowledge. They encompass psychological activities that acquire, store, retrieve, transform, or apply information. In cognitive robotics, we engineer machines to replicate or approximate these cognitive processes.

- **Perception:** Sensing and interpreting the environment (vision, sound, touch).
- **Memory:** Storing a map of the environment, past experiences, and learned knowledge.
- **Attention:** Focusing computational resources on the most relevant parts of the environment.
- **Reasoning:** Making logical inferences about the world and what actions to take.
- **Planning:** Deciding a sequence of actions to achieve a goal.
- **Learning:** Updating internal knowledge based on new experiences.

6.3 Core Issues in Cognitive Robotics

While traditional cognitive modelling approaches assumed symbolic coding schemes as a means of depicting the world, translating the real world into these kinds of symbolic representations has proven to be problematic — if not entirely unworkable for rich, real-world environments.

The two core issues that must be addressed:

- **Perception and Action:** How does the robot convert raw sensory data into meaningful representations, and then translate decisions back into physical actions?
- **Symbolic Representation:** How should knowledge about the world be encoded? Pure symbolic approaches (like classical AI) have proven too brittle. Modern cognitive robots use hybrid approaches combining symbols with sub-symbolic (neural network) representations.

6.4 Target Cognitive Capabilities

Cognitive robotics takes human or animal cognition as its starting point — opposed to traditional AI which starts from mathematical optimisation. The target robotic cognitive capabilities include:

- Perception processing — understanding raw sensory input.

- Attention allocation — deciding what to focus on in a complex scene.
- Anticipation — predicting what will happen next based on current state.
- Planning — deciding a sequence of actions to achieve a goal.
- Complex motor coordination — smooth, precise physical movement.
- Reasoning about other agents — understanding what other people or robots are doing and why.
- Mental state reasoning — potentially, reasoning about their own knowledge and uncertainty.

Robotic cognition embodies the behaviour of intelligent agents in the physical world. Ultimately, the robot must be able to act in the real world — not just in a simulation.

6.5 Key Aspects and Capabilities of Cognitive Robots

Learning and Adaptation

Cognitive robots utilise machine learning and developmental techniques (like developmental / epigenetic robotics) to improve their performance over time — often learning through experience, just as children develop motor and cognitive skills gradually.

Perception and Reasoning

Cognitive robots use AI to process sensory input, recognise objects, and reason about the best actions in uncertain situations. They do not just react — they reason about what they perceive.

Human-Robot Interaction (HRI)

A major focus of cognitive robotics is enabling robots to understand human social cues — gaze direction, gestures, facial expressions, and language — to collaborate effectively and naturally with humans.

Embodied Cognition

Intelligence in a cognitive robot is not just in the software — it is tightly coupled with the physical body and its interaction with the environment. The body's structure, sensors, and mechanics all shape what the robot can know and do.

Capability	Description	Example Application
Learning & Adaptation	Improves through ML and experience, not just programming	Robot learning to walk by trial and error
Perception & Reasoning	Processes sensors, recognises objects, reasons about actions	Identifying and sorting unknown objects
Human-Robot Interaction	Understands gestures, gaze, language for collaboration	Collaborative factory robot that works safely with humans
Embodied Cognition	Intelligence emerges from body-environment interaction	Robot adjusting grip force by feeling object texture

6.6 Soft Computing in Robot Cognition

Soft computing tools are the practical building blocks that enable cognitive capabilities in real robots:

- **Fuzzy Logic:** Handles the vagueness and imprecision of real-world sensor readings and natural language commands.
- **Neural Networks (ANN / CNN / RNN):** Power the perception, recognition, and learning capabilities of the robot.
- **Genetic Algorithms:** Optimise robot control policies, neural network architectures, and motion planning parameters.
- **Hybrid Neuro-Fuzzy:** Combines learning (ANN) with interpretable rules (fuzzy) — ideal for robot control where both adaptability and transparency are needed.

7. Quick Revision — Unit III at a Glance

7.1 All Topics — One-Line Summaries

- **Cognition:** Mental processes for acquiring, storing, retrieving, and applying knowledge.
- **Model of Cognition:** Computational/theoretical representation of mental processes used to simulate or predict human behaviour.
- **Cybernetic View:** Treats robots as feedback-controlled systems that sense → process → act → observe outcome, mimicking biological brains.
- **Visual Perception:** Machine capability to interpret images/video/3D scenes — object detection, segmentation, tracking, 3D reconstruction.
- **Visual Recognition:** Identifying/labelling specific elements in images — image classification, object detection, facial recognition, segmentation.
- **ML in Perception:** CNNs and deep learning are the standard tools for modern robot perception — replaces brittle hand-coded rules.
- **Soft Computing:** Techniques (Fuzzy, ANN, GA, SVM, Hybrid) that handle imprecision, uncertainty, and complexity to give 'good enough' solutions.
- **Fuzzy Logic:** Handles degrees of truth (0 to 1) — ideal for vague concepts like 'hot', 'fast', 'close'.
- **Genetic Algorithms:** Optimisation inspired by natural selection — population evolves via selection, crossover, mutation.
- **SVM:** Finds the maximum-margin hyperplane that best separates classes — effective for high-dimensional data.
- **Cognitive Robotics:** Engineering robots with human-like cognitive abilities: perception, reasoning, learning, planning, HRI.
- **Core Issues:** Perception-action loop and symbolic representation are the central challenges of cognitive robotics.
- **Embodied Cognition:** Intelligence arises from the interaction of the body with the environment — not just from the software.

7.2 Visual Perception vs Visual Recognition

Aspect	Visual Perception	Visual Recognition
Scope	Broad — interpreting the entire visual scene	Specific — labelling elements within a scene
Includes	Motion, depth, 3D structure, tracking	Classification, detection, facial ID, segmentation
Output	Rich scene understanding	Class labels, bounding boxes, masks
Relationship	Perception contains recognition as a subset	Recognition is one component of perception

7.3 Soft Computing Tools — Quick Reference

Tool	Handles	Key Mechanism	Best For
Fuzzy Logic	Imprecision, vague concepts	Degrees of truth (0–1), IF-THEN fuzzy rules	Control systems, vague decision rules
ANN	Complex patterns, non-linear data	Layers of neurons + backpropagation	Image, speech, pattern recognition
Genetic Algorithm	Optimisation, search	Selection + Crossover + Mutation	Scheduling, design optimisation
SVM	Classification, regression	Maximum-margin hyperplane	Text/image classification, small datasets
Neuro-Fuzzy (Hybrid)	Both learning and uncertainty	ANN learns fuzzy rules from data	Robot control, adaptive inference

7.4 Cognitive Robotics — Key Terms

Term	Meaning
Embodied Cognition	Intelligence is shaped by the physical body and its environment — not just the software.
Epigenetic Robotics	Robots that develop capabilities over time through interaction — inspired by child development.
HRI (Human-Robot Interaction)	Study/design of systems for natural, safe interaction between humans and robots.
Perception-Action Loop	Continuous cycle: sense → perceive → decide → act → receive feedback → repeat.
Symbolic Representation	Encoding world knowledge as symbols and rules — the traditional AI approach, but brittle in practice.
Sub-symbolic Representation	Encoding knowledge as distributed patterns in neural network weights — more robust for perception.
